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- 30 min
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🎉 Congratulations! You passed!

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Module 1 Quiz: Introduction, Probability, Expectations, and Random Vectors

1. What is $P(A \cup B)$ always equal to?

COOCH

👉 Correct

Ready to review what you've learned before taking the quiz? I'm here to help.

➡ Help me practice

➡ Let's chat

1 / 1 point

⌵

2. Which of the following are symmetry conditions for $f(x)$? (Check all that apply.)

Due Jul 24, 11:59 PM IST Attempts 3 every 8 hours

👉 Correct

👉 Receive grade

To Pass 80% or higher

Your grade 95%

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We try to present some

1 / 1 point

Try again

3. Consider influenza epidemics for two parent heterosexual families. Suppose that the probability is 17% that at least one of the parents has contracted the disease. The probability that the father has contracted influenza is 12% while the probability that both the mother and father have contracted the disease is 6%. What is the probability that the mother has contracted influenza?

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👉 Correct

1 / 1 point

4. A random variable, X , is uniform, so that it's density is $f(x) = 1$ for $0 \leq x \leq 1$. What is it's 75th percentile? Express your answer to two decimal places.

👉 Correct

1 / 1 point

5. A Pareto density is $\frac{1}{x^2}$ for $1 < x < \infty$. What is the distribution function associated with this density for $1 < x < \infty$?

👉 Correct

1 / 1 point

6. What is the quantile p from the density $e^{-x}(1 + e^{-x})^{-2}$?

👉 Correct

1 / 1 point

7. Suppose that a density is of the form cx^k for some constant $k > 1$ and $0 < x < 1$. What is the value of c ?

👉 Correct

1 / 1 point

8. Suppose that the time in days until hospital discharge for a certain patient population follows a density $f(x) = \frac{1}{2} \exp(-x/2)$ for $x > 0$. What is the median discharge time in days?

👉 Correct

1 / 1 point

9. Consider the density given by $2x \exp(-x^2)$ for $x > 0$. What is the median?

👉 Correct

1 / 1 point

10. Suppose $h(x)$ is such that $\infty > h(x) > 0$ for $x = 1, 2, \dots, I$. Then $c \times h(x)$ is a valid pmf when c is equal to what?

👉 Correct

1 / 1 point

11. Let $g(x) = \pi f_1(x) + (1 - \pi)f_2(x)$ where f_1 and f_2 are densities with associated means μ_1 and μ_2 , respectively. Here $0 \leq \pi \leq 1$. Note that g is a valid density. What is it's associated mean?

👉 Correct

1 / 1 point

12. Suppose that a density is of the form $(k + 1)x^k$ for some constant $k > 1$ and $0 \leq x \leq 1$. What is the mean associated with this density?

👉 Correct

1 / 1 point

13. You are playing a game with a friend where you flip a coin and if it comes up heads you give her X dollars and if it comes up tails she gives you Y dollars. The probability that the coin is heads is p (some number between 0 and 1.) What has to be true about X and Y to make so that both of your expected total earnings is 0. (The game would then be called "fair".)

👉 Correct

1 / 1 point

14. You are playing a game with a friend where you flip a coin and if it comes up heads you give her 1 dollar and if it comes up tails she gives you one dollar. If you play the game 10 times what would be the variance of your earnings?

👉 Correct

1 / 1 point

15. When at the free-throw line for two shots, a basketball player makes at least one free throw 90% of the time. 80% of the time, the player makes the first shot, while 70% of the time she makes both shots. Which number is closest to the conditional probability that the player makes the second shot given that she missed the first?

👉 Correct

1 / 1 point

16. Let $X_1, \dots, X_n$ be random variables independent of $Y_1, \dots, Y_m$, where both groups are iid with associated population means μ_1 and μ_2 and population variances σ_1^2 and σ_2^2 , respectively. Let $\bar{X}$ and $\bar{Y}$ be their sample means. What is the variance of $\bar{X} - \bar{Y}$?

👉 Correct

1 / 1 point

17. Quality control experts estimate that the time (in years) until a specific electronic part from an assembly line fails follows (a specific instance of) the Pareto density, $f(x) = \frac{2}{x^3}$ for $1 < x < \infty$. Which option is closest to the mean failure time?

👉 Correct

1 / 1 point

18. Let f be a continuous density having a finite mean and μ be any number. Suppose that $f(x) = f(-x)$ (i.e. f is symmetric about 0). Convince yourself that $f(x - \mu)$ is a valid density. What is its associated mean?

👉 Incorrect

0 / 1 point

19. Suppose that f be a mean 0 density having variance 1. What is the variance associated with the density $g(x) = f(x/\sigma)/\sigma$?

👉 Correct

1 / 1 point

20. If X and Y are mean 0, variance 1 independent random variables, what is $E[X^2Y^2]$?

👉 Correct

1 / 1 point